

● MEDIUM

● ALGEBRA

● ANSWER KEY

Linear functions

06

3 answers · Rationale-rich · Teach from doc

Q1**774**

The correct answer is 774. It's given that Brian saves $\frac{2}{5}$ of the \$ 215 he earns each week from his job. Therefore, Brian saves $\frac{2}{5}$ \$ 215, or \$ 86, per week. If Brian continues to save at this rate of \$ 86 per week for 9 weeks, then he will save a total of 986, or 774, dollars.

Q2**A**

A. The electrician's hourly rate

Choice A is correct. It's given that the electrician charges a onetime fee plus an hourly rate. It's also given that the graph represents the total charge, in dollars, for x hours of work. This graph shows a linear relationship in the xy -plane. Thus, the total charge y , in dollars, for x hours of work can be represented as $y = mx + b$, where m is the slope and b is the y -intercept of the graph of the equation in the xy -plane. Since the given graph represents the total charge, in dollars, by an electrician for x hours of work, it follows that its slope is m , or the electrician's hourly rate.

Choice B is incorrect. The electrician's onetime fee is represented by the y -coordinate of the y -intercept, not the slope, of the graph.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Q3**11**

The correct answer is 11. It's given that fx is defined by the equation $f(x) = mx - 28$, where m is a constant. It's also given in the table that when $x = 10$, $fx = 82$. Substituting 10 for x and 82 for fx in the equation $fx = mx - 28$ yields, $82 = m10 - 28$. Adding 28 to both sides of this equation yields $110 = 10m$. Dividing both sides of this equation by 10 yields $11 = m$. Therefore, the value of m is 11.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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